

Design and Analysis of Single-Inductor Power Converter for Both Battery Balancing and Voltage Regulation

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Abstract—In this paper, a highly efficient battery-balancing circuit with output voltage regulation for mobile applications is proposed and analyzed. The proposed circuit performs a cell balancing operation for series-connected multiple battery cells and voltage regulation operation simultaneously without additional power conversion stages. Therefore, it can be integrated into a battery pack with maximized power density and efficiency. These features are essential for recent low-power applications with multiple batteries, such as wireless/smart speakers, drones, and electronic point of sale that require high power density. The operational principle, electrical modeling, and trade-off relationship between system power losses and balancing characteristics are derived, and the optimal design candidates of the proposed topology are discussed in terms of Pareto optimization. The effectiveness of the proposed system was verified with a prototype board using GaN switches, and 96.05% efficiency was achieved with a 1.8 V, 0.5 A load.

Index Terms—Battery management systems, battery-balancing circuits, power conversion circuits, output voltage regulation

NOMENCLATURE

$S_{S1,T} - S_{S3,T}$	Top switch for each cell selection
$S_{S1,B} - S_{S3,B}$	Bidirectional bottom switch for each cell selection
$S_{P1} - S_{P3}$	High-side switch for each cell
S_L	Low-side switch
V_P	Pole voltage with respect to battery ground
V_O	Output differential voltage
i_{L1}, i_L	Output inductor, load current
I_S, I_T	Strong cell, target cell current
V_S, V_T	Strong cell, target cell voltage
$i_{BAL,pk}, i_{BAL,avg}$	Peak value, average value of balancing current
t_{C1}, t_{L1}, t_{P1}	Time for high-side on, low-side on, and a sum of both in Mode 1
t_{C2}, t_{L2}, t_{P2}	Time for high-side on, low-side on, and a sum of both in Mode 2

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D_{M1}, D_{M2}	Time duration of Mode 1 and 2
$SOC_S, \Delta SOC_S$	Source cell SOC and its variance during balancing operation
$SOC_T, \Delta SOC_T$	Target cell SOC and its variance during balancing operation
SOC_i	Intermediate cell SOC
P_L, P_T, P_S	The load power, target cell power, and source cell power
M	The conversion ratio between the source cell V_S and the output voltage V_O .
R_{BAT}, R_{SW}	The internal resistance of the single battery cell, on-resistance of the switch.
$P_{BAT,1}, P_{COND,1}$	The conduction loss due to battery internal resistance, switch resistance in Mode 1
$P_{OV,1}, P_{SW,1}$	The overlap loss, switching loss in Mode 1
$P_{BAT,2}, P_{COND,2}$	The conduction loss due to battery internal resistance, switch resistance in Mode 2
$P_{OV,2}, P_{SW,2}$	The overlap loss, switching loss in Mode 2

I. INTRODUCTION

With the rapid development of mobile, wearable devices, and other commercial battery-powered applications, demand for secondary batteries is on the rise. In particular, a battery-balancing circuit is essential for systems with multiple batteries connected in series since it contributes to maximizing the utilization capacity of the battery energy. Typically, battery balancing is performed using either passive balancing or active balancing [1]–[4]. For passive balancing, excess energy is dissipated to heat with a pair of shunt connected switch and resistor. In order to reduce loss during balancing operation, an active balancing method is proposed [2], [3], [5]–[7]. It transfers energy between cells using switches and passive components such as capacitors, inductors, and transformers. Active balancing methods are classified into several categories depending on the components. The switched capacitor (SC) method [8] is advantageous for integration into a system because no magnetic components are used. However, the equalization speed is relatively slow compared to that of methods using magnetic components [9]–[11]. Balancing methods using an inductor or transformer are widely used since

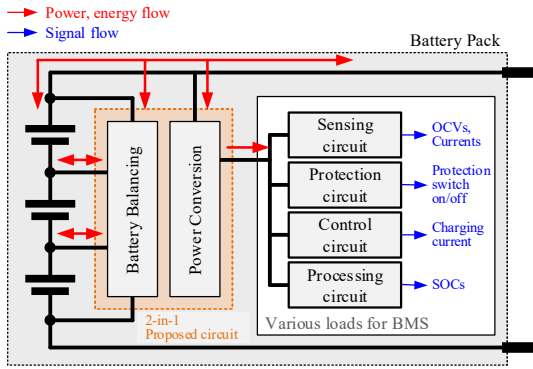


Fig. 1. Typical power, energy, and signal flow of a battery pack consisting of multicell batteries and a battery management system (BMS) [18]. The electrical requirements of the power conversion circuit are increasing [19], which limits the power density of the entire battery-powered system. In this work, battery balancing and voltage regulation circuits are combined into a single circuit.

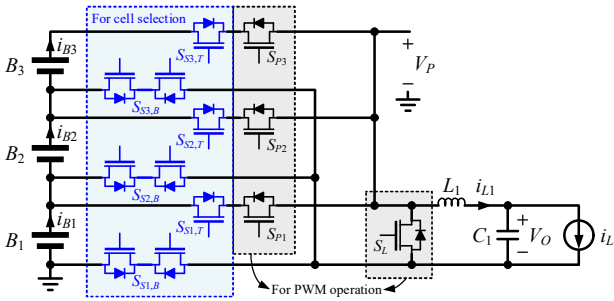


Fig. 2. The proposed battery-balancing circuit with voltage regulation for three series-connected battery cells. In this study, three cells are used, and the analysis and operation performed in this study are all applicable to a system with more cells.

the equalization speed is fast and the efficiency is high. However, the occupied area is relatively large and the control is complex. Also, when using multi winding transformers, the design of leakage inductance with high accuracy of the secondary winding is required, and the overall system becomes complicated and the size is increased [3], [10].

Recently, in many emerging applications such as wireless/smart speakers, drones, and electronic point-of-sale (EPOS) [12]–[14], the use of multiple batteries is on the rise to shorten the charging time or due to the increase of loads (display, printing, low power motor drive, etc.) that require a high supply voltage [12], [14]–[16].

Fig. 1 shows a typical power, energy, and signal flow of a battery pack consisting of multicell batteries and its battery management system (BMS) [17]. A BMS has many features as shown in Fig. 1, thus a power conversion circuit for various functions is required to properly operate a BMS. Since the aforementioned low-power emerging applications require small volumes and low weights, the power conversion circuits limit the power density due to magnetic components [15], [18]. Therefore, the optimal design of battery-balancing circuits and power conversion circuits is a key for achieving a high-performance BMS [19].

For this reason, as shown in Fig. 1, there have been studies

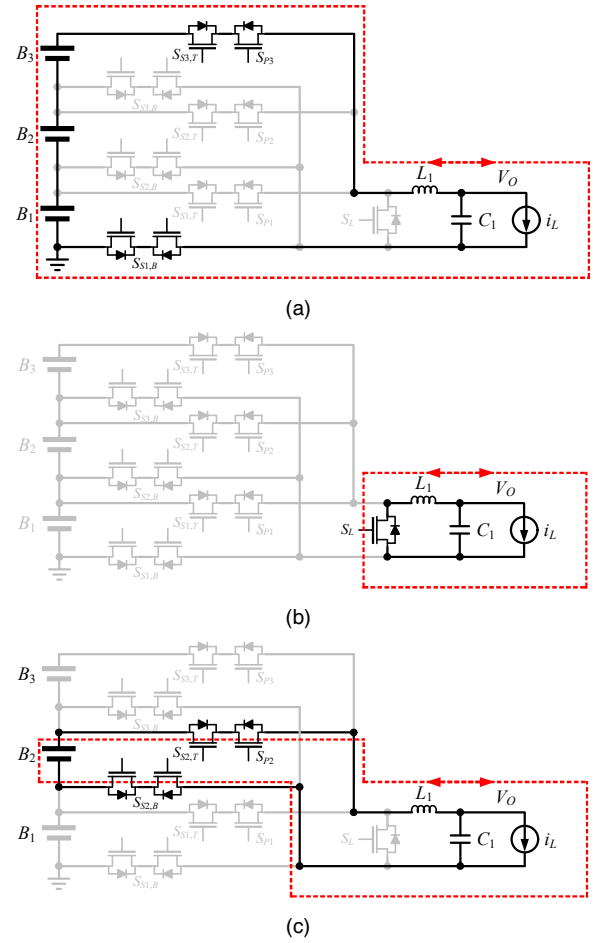


Fig. 3. The operation mode of the proposed topology: (a) pack-powering mode, (b) load-powering mode, and (c) cell-powering mode; in this operation mode, for example, B_2 is selected.

that have achieved both battery-balancing operations and power conversion (output voltage regulation) [19]–[22]. However, to the best of our knowledge, previous studies have focused on functional implementation rather than studies that provide in-depth analysis and optimized design. Previous related studies have the following limitations and disadvantages.

- 1) Due to the lack of an analytical approach, similar performance and efficiency are not guaranteed for the different operating conditions [19]–[22].
- 2) Detailed analysis of the balancing operation is not considered [22].
- 3) Implementation of functions (battery balancing and power conversion) is only covered without study or interpretation on how the performance (efficiency, balancing speed) is achieved and optimized [19].
- 4) Transformer type topology (flyback) limits the power density, especially in low-power, low-voltage applications [19].

As described above, previous studies focused on the implementation of the functions by relying on the conventional balancing performance. This paper provides analysis, trade-off relationships, and system optimization of the single-inductor power converter for both battery balancing and voltage

regulation through the consideration of balancing time, balancing current, and power loss. To perform both battery balancing and voltage regulation, the circuit shown in Fig. 2 is considered for a low-power multiple-battery system. The topology, with a single-inductor power converter, is a good candidate to achieve high efficiency and fast balancing speed with a small area/size for low-power applications. This study is an extension of a previous conference paper [23] that introduces the optimal design of the proposed circuit.

In the proposed topology (Fig. 2), bidirectional switches are used to select a cell to transfer energy back and forth. The load side consists of an inductor L_1 and capacitor C_1 that supply power to the load and deliver energy to the target cell (weak cell). Detailed descriptions are covered in Section II. In Section III, the balancing characteristics with power conversion are modeled, and the relationship between the balancing time and state of charge (SOC) value versus the balancing current is shown. The loss modeling of the proposed circuit is discussed in Section IV, and the optimal design of the proposed circuit is performed in Section V with the loss-balancing time Pareto optimization. In Section VI, the three design candidates are verified with a prototype board and then compared. Finally, conclusions are drawn, including discussions of previous works and experimental highlights.

II. OPERATIONAL PRINCIPLE

In this section, the operational principle of the proposed topology is described. As shown in Fig. 3, the proposed circuit has three operating modes depending on where the equalization current flows. In pack-powering mode, as shown in Fig. 3(a), the current flows from the pack to the load, or vice versa. In load-powering mode, the transferred energy is used in the load or reserved in the load capacitor C_1 for delivering back to the cell or pack, as shown in Fig. 3(b). In cell-powering mode, as shown in Fig. 3(c), the current flows from the cell to the load, or vice versa. In each mode, the direction of the current is determined via appropriate control of switches.

As shown in Fig. 3, the proposed topology provides three types of balancing operations: pack-to-cell (P2C), cell-to-cell (C2C), and cell-to-pack (C2P). Among three possible operations, studies on balancing performance have shown that C2C-type, particularly direct-C2C-type, has faster equalization speeds than other types [2], [7].

From the perspective of power conversion, there are two types of conversion depending on the power input: pack or cell. Considering conduction loss I^2R , a pack-input has a larger resistance than a cell-input because the pack is composed of series-connected cells. Additionally, as shown in Fig. 4, for all ranges of battery voltages, the current ripple of the cell-input is less than that of the pack-input. A smaller current ripple yields a smaller RMS current, which means that the power loss for the cell type is smaller.

Thus, in terms of balancing speed and power conversion efficiency, it can be understood that the cell-input is more advantageous for efficient balancing and power conversion than the pack-input.

In this study, battery balancing and power conversion operations are performed through direct C2C operation, a

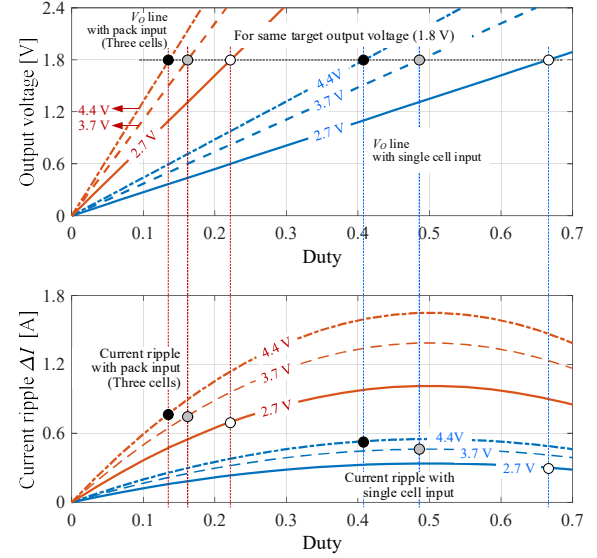


Fig. 4. The output voltage and current ripple according to the duty ratio according to input type for different battery voltages. With the same output voltage, same inductor, and same operating frequency, the cell-input is advantageous for high-efficiency due to the smaller current ripple. For pack input, three cells are assumed.

method specifically referred to as cell-to-load-to-cell (C2L2C) operation. For C2L2C operation, the load-powering mode (Fig. 3(b)) and cell-powering mode (Fig. 3(c)) are applied.

Typical waveforms of the C2L2C operation for battery balancing and voltage regulation of the proposed topology are shown in Fig. 5. For the following description, the direction from the battery to the load is considered to be the positive direction of the current, as shown in Fig. 2. V_O is well regulated with a voltage ripple ΔV , which is modified from a typical burst mode operation [24]. The battery voltage is constant during each mode, and the peak value of the inductor current i_{L1} is well regulated with $i_{BAL,pk}$.

Fig. 5 shows that the C2L2C operation is divided into two modes: Mode 1 and Mode 2, the cell-to-load (C2L) mode and load-to-cell (L2C) mode, respectively. The circuit operates in Mode 1 until the output voltage V_O reaches the high threshold; then, it remains in Mode 2 until the low threshold is reached. In the case of Mode 1, power is transferred from the source cell to the load with a positive inductor current i_{L1} , and in Mode 2, power is transferred from the load to the target cell with a negative inductor current i_{L1} .

The operating period of Mode 1 t_{p1} is expressed as

$$t_{p1} = t_{C1} + t_{L1} = L_1 \frac{i_{BAL,pk}}{V_S - V_O} + L_1 \frac{i_{BAL,pk}}{V_O}, \quad (1)$$

Likewise, the operating period of Mode 2 t_{p2} is expressed as follows.

$$t_{p2} = t_{L2} + t_{C2} = L_1 \frac{i_{BAL,pk}}{V_O} + L_1 \frac{i_{BAL,pk}}{V_T - V_O} \quad (2)$$

Notably, the peak value of the inductor current $i_{BAL,pk}$ is an important design element of the proposed circuit that is discussed in detail in the following section.

The change of output voltage in each mode is assumed to be linear, and the voltage changes in both modes are the same in

the steady-state; therefore, the time duration of each mode is expressed as follows.

$$D_{M1} \equiv \frac{t_1 - t_0}{t_2 - t_0} = \frac{0.5i_{BAL,pk} + i_L}{i_{BAL,pk}} \quad (3)$$

$$D_{M2} \equiv \frac{t_2 - t_1}{t_2 - t_0} = \frac{0.5i_{BAL,pk} - i_L}{i_{BAL,pk}} \quad (4)$$

III. BALANCING CHARACTERISTICS

In this section, the balancing characteristics of the proposed topology according to the balancing current and load current are described. The balancing characteristics, including the final SOC value and the balancing time of the proposed circuit, are considered; then, the power loss model is covered in the following section. The relationship between battery voltage and SOC is assumed to be a simple linear model for providing insights into the balancing characteristics of the proposed topology.

Fig. 6 shows the SOC plot for a typical C2L2C operation of the proposed topology. First, the source cell delivers energy to the load side. Then, the load side consists of the inductor L_1 , and capacitor C_1 delivers the energy to the target cell. In this operation, cells with the highest SOC and the lowest SOC are selected as the source cell and the target cell, respectively. Then, the charge/discharge variance for time t of each cell is as follows.

$$\Delta SOC_S(t) \cdot Q_{BAT} = \int_0^t I_S(t) dt = i_{BAL,avg} \cdot t \quad (5)$$

$$\Delta SOC_T(t) \cdot Q_{BAT} = \alpha \int_0^t \{\mu_B i_{BAL,avg} - M i_L(t)\} dt \quad (6)$$

Here, μ_B is the efficiency of the proposed circuit, which is $\mu_B = (P_L + P_T)/P_S$, where P_L , P_T , and P_S are the load power, target cell power, and source cell power, respectively. μ_B in (6) according to battery voltage is calculated by (14)-(23). M is the conversion ratio between the source cell V_S and the output voltage V_O , $M = V_O/V_S$. α is the ratio between the source cell V_S and the target cell V_T , $\alpha V_T = V_S$. Q_{BAT} is the capacity of the battery.

Considering three or more cells, the balancing of the highest or lowest SOC converges to the same SOC as the intermediate cell $SOC_i(0)$. In other words, t_x is obtained from the equations below.

$$\begin{cases} SOC_i(0) = SOC_S(0) - \Delta SOC_S(t_{x,a}) \\ SOC_i(0) = SOC_T(0) + \Delta SOC_T(t_{x,b}) \\ t_x = \min(t_{x,a}, t_{x,b}) \end{cases} \quad (7)$$

In Fig. 6(a), since the minimum value of t_x is determined by the source cell, t_x is equal to t_{x1} . Likewise, in Fig. 6(b), t_x is determined by the target cell, so t_x is equal to t_{x2} . In other words, the total balancing time t_{BAL} is divided into before and after the conditions in (7) are satisfied, i.e., t_{x1} & t_{y1} or t_{x2} & t_{y2} , as shown in Fig. 6. Before satisfying the conditions in (7), each cell is balanced with the rate of change of (5) and (6). After the condition is satisfied, the balancing rate is changed according to the relationship between each cell.

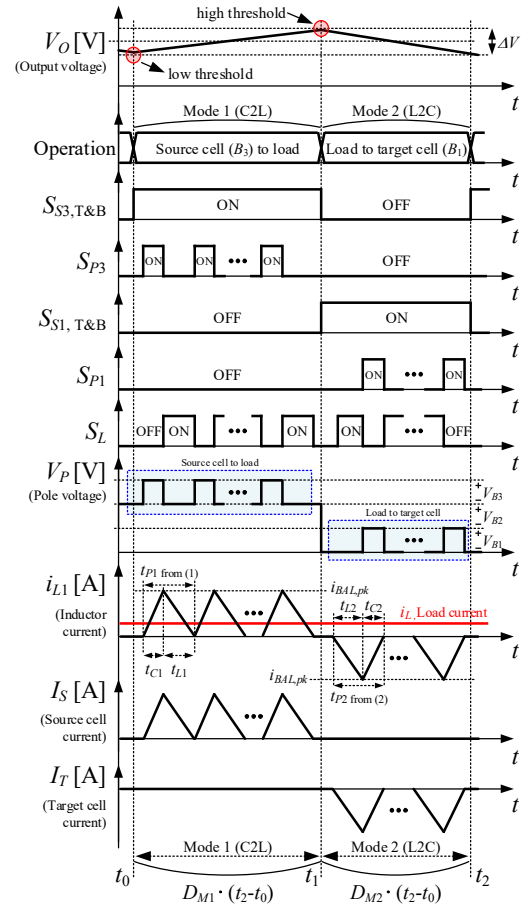


Fig. 5. Operational waveforms of the battery balancing with voltage regulation for C2L2C operation. In this waveform, B_3 is a source cell, and B_1 is a target cell. Mode 1 operates by repeating the pair of cell-powering mode (Fig. 3(c), B_3 in this waveform) and load-powering mode. Mode 2 operates by repeating the pair of load-powering mode and cell-powering mode (Fig. 3(c), B_1 in this waveform). t_{C1} , t_{L1} , t_{C2} , and t_{L2} follow (1) and (2).

For Case I of Fig. 6(a), when $i_{BAL}(t)$ and $i_L(t)$ are constant within the time period, the total balancing time is expressed as follows.

$$t_{BAL1} = t_{x1} + t_{y1} \quad (8)$$

$$t_{x1} = \frac{(SOC_S(0) - SOC_i(0)) \cdot Q_{BAT}}{i_{BAL,avg}} \quad (9)$$

$$t_{y1} = \frac{SOC_S(t_{x1}) - SOC_T(t_{x1})}{0.5i_{BAL,avg} + \alpha\{\mu_B i_{BAL,avg} - M i_L\}} \cdot Q_{BAT} \quad (10)$$

where t_{x1} and t_{y1} are the duration of time of each, as shown in Fig. 6(a). Likewise, the total balancing time for Case II is expressed as

$$t_{BAL2} = t_{x2} + t_{y2}, \quad (11)$$

$$t_{x2} = \frac{(SOC_i(0) - SOC_T(0)) \cdot Q_{BAT}}{\alpha\{\mu_B i_{BAL,avg} - M i_L\}} \quad (12)$$

$$t_{y2} = \frac{SOC_S(t_{x2}) - SOC_T(t_{x2})}{i_{BAL,avg} + 0.5\alpha\{\mu_B i_{BAL,avg} - M i_L\}} \cdot Q_{BAT} \quad (13)$$

As shown in Fig. 6, the SOC value at t_{BAL} is determined by both the balancing current and the load current. Considering the

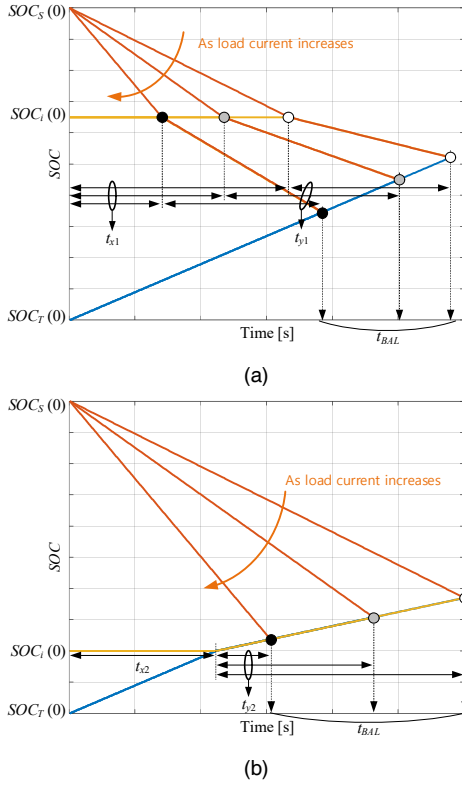


Fig. 6. Each SOC profile of the proposed topology (C2L2C operation) versus time is divided into two cases according to the conditions in (7). (a) Case I: the SOC of an intermediate cell is between the source cell SOC and the final SOC, (b) Case II: the SOC of an intermediate cell is between the final SOC and the target cell SOC.

two extreme cases, if the balancing current is very large compared to the load current, the final SOC (SOC_{FIN}) is the average value of each initial SOC ($\sum SOC(0)/3$), which is the achievable maximum value of the final SOC. On the other hand, if the balancing current is very small compared to the load current, the final SOC converges to $SOC_T(0)$, which is the minimum of the final SOC values.

Fig. 7 shows a plot of the final SOC value and balancing time with respect to balancing and load currents for operating conditions to be used in this study. The final SOC value tends to increase as the balancing current increases and tends to decrease as the load current increases. However, the balancing time becomes shorter as both currents increase.

IV. LOSS MODELING

This section describes the loss modeling of the proposed circuit. The losses of each mode are divided into 1) battery conduction loss due to the internal resistance of the battery cell, 2) switch conduction loss due to the on-resistance of the power switch, 3) switch overlap loss, and 4) switching loss due to the output capacitance of each switch. Modeling for each loss factor is derived based on the operational principle described in Section II, and the time-average value of each loss component is expressed to determine the total loss.

Before proceeding with the loss analysis, the detailed switch operation of C2L2C is described. For both Modes 1 and 2, the operation is performed with four high-side switches and one

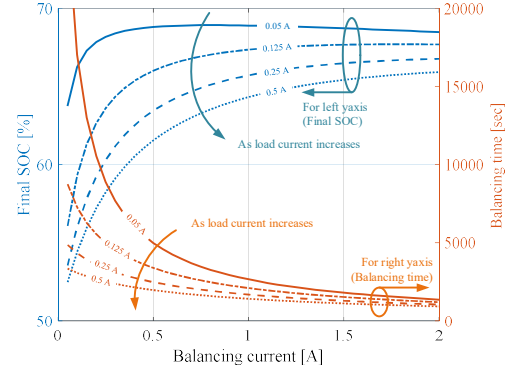


Fig. 7. Plot of the final SOC and balancing time as a function of load current and balancing current; the final SOC increases with an increase in the balancing current and tends to decrease with an increase in the load current. The balancing time tends to decrease with increasing balancing current and increasing load current. The initial SOC values are the same as the values of the experiment (Fig. 17).

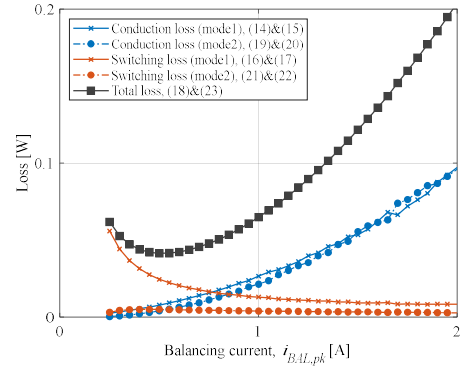


Fig. 8. Plot of the total power loss versus the balancing current with $4.7 \mu\text{H}$ of inductance; the switching losses are dominant as the balancing current decreases, and the conduction losses are dominant as the balancing current increases.

low-side switch S_L , as shown in Fig. 2. For example, three of the four high-side switches S_{S1T} and S_{S1B} (S_{S1B} represents two switches) are responsible for cell selection, and the other S_{P1} is responsible for PWM switching paired with the low-side switch S_L for B_1 .

Thus, the battery conduction loss for Mode 1 is expressed as

$$P_{BAT,1} = i_{BAL,rms}^2 R_{BAT} = \left(i_{BAL,pk} \sqrt{\frac{t_{C1}}{3t_{P1}}} \right)^2 R_{BAT}, \quad (14)$$

where R_{BAT} is the internal resistance of the single battery cell. The switch conduction loss for Mode 1 is expressed as

$$P_{COND,1} = \left[4 \left(i_{BAL,pk} \sqrt{\frac{t_{C1}}{3t_{P1}}} \right)^2 + \left(i_{BAL,pk} \sqrt{\frac{t_{L1}}{3t_{P1}}} \right)^2 \right] R_{SW}, \quad (15)$$

where R_{SW} is the on-resistance of the switch. Since the battery cell transition period is longer than the PWM switching period, i.e., $t_{P1} \ll (t_2 - t_0)$, the PWM pairs ($S_{P1} \sim S_{P3}$, and S_L) form dominant switching losses. Thus, the switch overlap loss and switching loss for Mode 1 are expressed as

$$P_{OV,1} = 2 \frac{0.5 i_{BAL,pk} V_S}{t_{P1}} T_{tr}, \quad (16)$$

$$P_{SW,1} = 2 \frac{0.5 C_{OSS} V_S^2}{t_{P1}}, \quad (17)$$

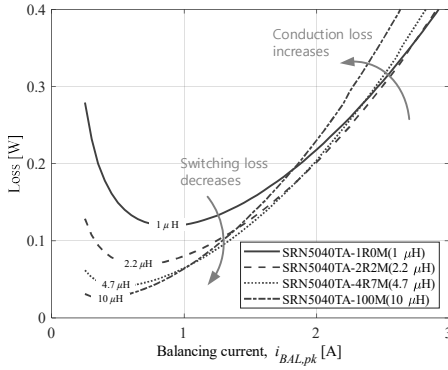


Fig. 9. Plot of the total power loss versus various inductances.

where V_S is the voltage of the source cell, T_{tr} is the ON/OFF transition time of each switch, and C_{OSS} is the output capacitance provided from the datasheet of the component.

Therefore, the total power loss for Mode 1 is expressed as follows.

$$P_1 = (P_{BAT,1} + P_{COND,1} + P_{OV,1} + P_{SW,1})D_{M1} \quad (18)$$

In a similar manner, the power loss for Mode 2 is expressed as follows.

$$P_{BAT,2} = i_{BAL,rms}^2 R_{BAT} = \left(i_{BAL,pk} \sqrt{\frac{t_{L2}}{3t_{P2}}} \right)^2 R_{BAT}, \quad (19)$$

$$P_{COND,2} = \left[4 \left(i_{BAL,pk} \sqrt{\frac{t_{L2}}{3t_{P2}}} \right)^2 + \left(i_{BAL,pk} \sqrt{\frac{t_{C2}}{3t_{P2}}} \right)^2 \right] R_{SW}, \quad (20)$$

$$P_{OV,2} = 2 \frac{0.5 i_{BAL,pk} V_T}{t_{P2}} T_{tr}, \quad (21)$$

$$P_{SW,2} = 2 \frac{0.5 C_{OSS} V_T^2}{t_{P2}}, \quad (22)$$

$$P_2 = (P_{BAT,2} + P_{COND,2} + P_{OV,2} + P_{SW,2})D_{M2}. \quad (23)$$

The total loss $P_1 + P_2$ versus the balancing current is illustrated in Fig. 8. The conduction loss represents the sum of P_{BAT} and P_{COND} , and the switching loss represents the sum of P_{OV} and P_{SW} . As in (1) and (2), the operating frequency and duty ratio are closely related to $i_{BAL,pk}$. In other words, the trend of power loss shown in Fig. 8 includes the tendency of not only conduction loss but also switching loss according to $i_{BAL,pk}$. It is observed that the switching losses become dominant as the balancing current decreases, the conduction losses become dominant as the balancing current increases. Thus, the balancing current $i_{BAL,pk}$ directly determines the total power loss, which is an important factor in the overall design process.

V. PARAMETER DESIGN AND SYSTEM OPTIMIZATION

In this section, the effect of each design consideration on the overall performance is investigated, and optimal design candidates are extracted via system optimization.

A. Output inductor and capacitor design

As shown in (1) and (2), the operating frequency of the proposed circuit is directly affected by the output inductance L_1 . Specifically, the change in inductance is closely related to the switching loss. Since the proposed circuit is integrated into the

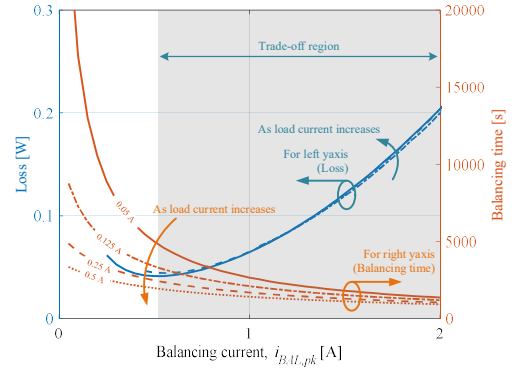


Fig. 10. Plot of the trade-off region between power loss and balancing time for various balancing currents and load currents (4.7 μ H of inductance). The initial SOC values are the same as the values of the experiment (Fig. 17).

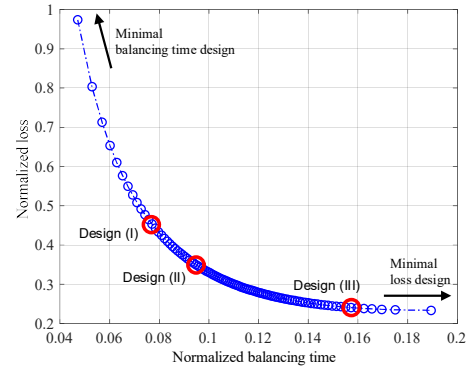


Fig. 11. The Pareto Front of the proposed circuit and three selected design candidates.

battery pack of mobile and consumer devices, such as wireless/smart speakers, drones, and EPOS, the output inductor must be small in size with small low-frequency copper loss. In this study, a set of commercial inductors with the same package size is selected [25]. For a given package size, an increase in inductance would result in a larger low-frequency copper loss since only the number of coil turns increases. As shown in Fig. 9, an increase in inductance L_1 reduces the switching loss and increases the conduction loss, as mentioned above. In this study, 4.7 μ H of inductance was selected considering both the size and power loss of the output inductor for the desired operation of the proposed circuit.

For the output capacitor C_1 , it is required to transfer the energy to the target cell from the load side. Thus, the output capacitor C_1 is designed to satisfy the following condition.

$$C_1 > L_1 i_{BAL,pk}^2 / \Delta V^2 \quad (24)$$

Here, the ripple of the voltage ΔV and the capacitor have a trade-off relationship. As the size of the capacitor C_1 increases, the ripple of the voltage is reduced. Thus, ΔV and C_1 are required by the given specification of a system.

B. Effect of balancing current

Based on the modeling analyzed in the previous section, the current-dependent loss and balancing time are shown in Fig. 10. Despite the change in the load current, the power loss remains

TABLE I
PARAMETER VALUES SUMMARY

Description	Values
Batteries	Li-ion 18650 2000 mAh
Switches	EPC2030
Gate drivers	Analog Devices - LTC4440
Output inductor L_1	Bourns - SRN5040TA-4R7M,
	4.7 μ H
Output capacitor C_1	MLCC 6.3 V Rating
	296 μ F (47 μ F * 5 + 10 μ F * 5 + 2.2 μ F * 5)
Output voltage V_O	1.8 V
Switching frequency	120 ~300 kHz
Rated output load current I_L	0.5 A

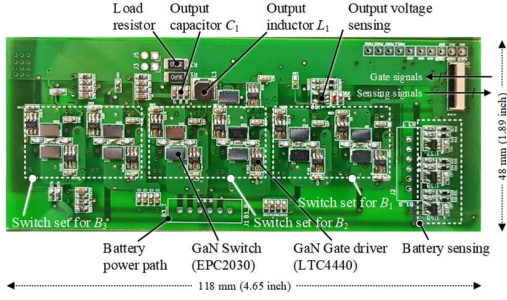


Fig. 12. A photograph of the prototype board of the proposed circuit with a three series-connected battery string.

at a similar level because the duty ratios in Mode 1 and Mode 2 (D_{M1} and D_{M2}) have the opposite direction to the load current I_L , which is observed in (3) and (4). Therefore, the total loss does not change significantly by the load current.

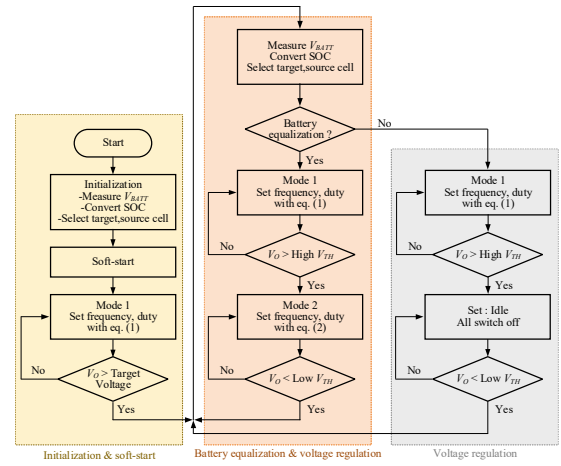
The relationship between the loss and balancing current is illustrated in Fig. 8 in the previous section. As observed in Fig. 7, the balancing time decreases as the balancing current (or load current) increases. In other words, when operating in the gray zone of Fig. 10 with designed values, a trade-off relationship between the power loss and balancing performance is inevitable. The parameter values used in this analysis are shown in Table I.

C. Design optimization with the Pareto Front

The design parameters are selected to optimize both the balancing speed and system efficiency. Design factors such as the balancing current, load current, output inductance, output capacitance, operating frequency, duty ratio, and output voltage ripple are taken into account for the optimal design of the proposed circuit. Based on the analysis performed in the previous section, the balancing time and total power loss are optimized by exploring the Pareto Front [26], [27]. The Pareto Front is the set of feasible optimum outputs. The equations done in Sections III & IV are coded and explored by using the *fgoalattain* function in the MATLAB optimization toolbox [28]. In this work, each parameter is normalized to its maximum value in the operating area with the design constraints.

In Fig. 11, the Pareto Front illustrates the trade-off relationship observed in Fig. 10. Considering the safe region of the battery used in this work, three design candidates are selected to validate this optimization process, and the balancing current $i_{BAL,pk}$ for designs I, II, and III is 1.25 A, 1.05 A, and 0.65 A, respectively.

VI. EXPERIMENTAL RESULTS



Note: High V_{TH} and low V_{TH} are the high threshold and low threshold of V_O as shown in Fig. 5, respectively.

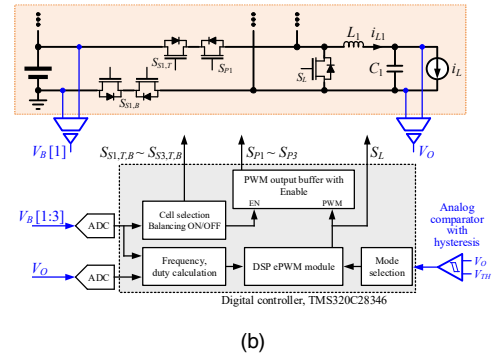


Fig. 13. (a) Flowchart of the proposed operation, (b) system configuration of the proposed circuit

The prototype board for verifying the proposed circuit and the analysis performed in this study is shown in Fig. 12. In this study, GaN switches and gate drivers are used to minimize the power loss and component area. The board consists of a bidirectional switch set, output inductor L_1 , output capacitor C_1 , and sensing circuit. The component values used in the prototype board are summarized in Table I.

Fig. 13(a) shows the flowchart of the proposed operation. At the start of the operation, the PWM duty ratio is set to zero and slowly increases until the target voltage is reached with the source cell. When the output voltage reaches the target value, the controller determines whether the balancing operation is required by sensing the battery voltages. If the balancing operation is required, the operating frequency and duty ratio for each mode are calculated from (1) and (2) to operate with targeted $i_{BAL,pk}$. If the tolerance is smaller than a certain value, only voltage regulation is performed without a balancing operation via cell-to-load (C2L) power conversion. Since the balancing current $i_{BAL,pk}$ is calculated by (1) and (2), the battery and output voltage are required to be sensed as shown in Fig. 13(b). The output voltage is sensed and detected with an analog comparator as shown in Fig. 13(b). Each signal for cell selection and PWM are properly generated according to the battery voltages by a digital controller (TMS320C28346, TI).

The C2L2C operation is shown in Fig. 14. The output voltage V_O stably reaches the target voltage with the source cell.

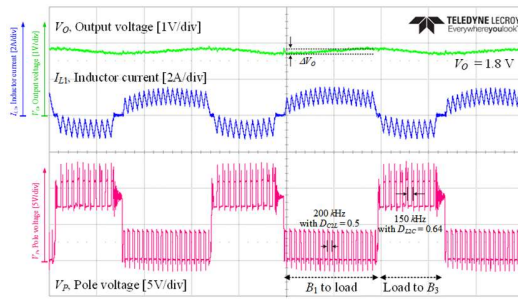


Fig. 14. The C2L2C operation of the proposed circuit. As observed, the output voltage operates in burst mode, as described in the previous section, and repeats the C2L mode (B_1 to load) and L2C mode (load to B_3) within the voltage window. The initial SOC of each cell is as follows: $SOC_1=90\%$, $SOC_2=65\%$, and $SOC_3=40\%$.

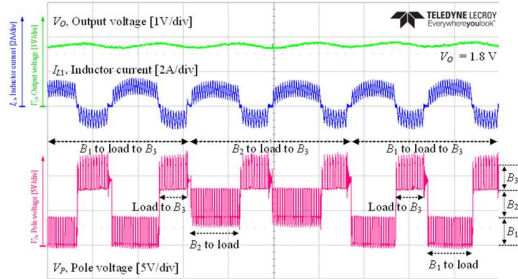


Fig. 15. The waveforms of output voltage V_O , pole voltage V_P , and inductor current I_{L1} for the C2L2C mode of the proposed circuit; two high SOC cells (B_1 and B_2) have similar values as the balancing operation proceeds. B_1 and B_2 alternately perform as the source cell and transfer energy to B_3 . The output load voltage V_O is stable despite cell changes.

For output voltage regulation, the burst mode with a transition between Modes 1 and 2 is implemented. The operation frequency and its duty ratio follow the relations in (1)-(4). In Mode 1, B_1 is selected as the source cell, and energy is transferred to the load. In Mode 2, the energy stored in load capacitor C_1 is delivered through inductor L_1 to target cell B_3 . The mode transition can be observed through the change in the pole voltage V_P and direction of the inductor current i_{L1} , as shown in Fig. 5.

Fig. 15 shows that two high energy cells (B_1 and B_2) have similar values as the balancing operation proceeds. B_1 and B_2 alternately perform as source cells and transfer energy to B_3 . The output voltage V_O is well regulated despite cell changes. Additionally, the operation can be observed through the pole voltage V_P and the inductor current i_{L1} .

The high-frequency oscillations at the end period as shown in Fig. 14 and Fig. 15. In this study, corresponding power loss is minimized through the proper selection of GaN switches and the minimum parasitic inductance PCB design [29].

Fig. 16 shows the measured SOC over time under an unequal initial SOC distribution. The experimental results are compared with the MATLAB estimated results based on (5)-(13). The overall balancing performance indicates adequate accuracy between the estimated and measured results.

Fig. 17 shows the measured efficiency and balancing time for each design candidate. As explained in the previous section, Design I has the shortest balancing time, and Design III has the highest efficiency. The validity of the analysis and optimization

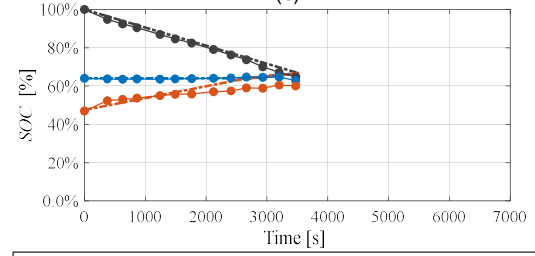
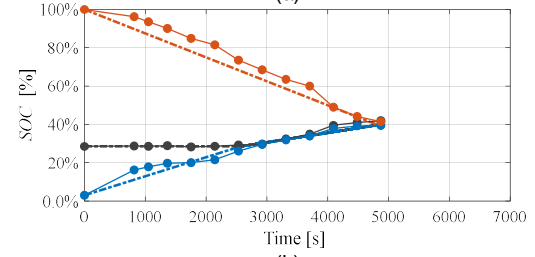
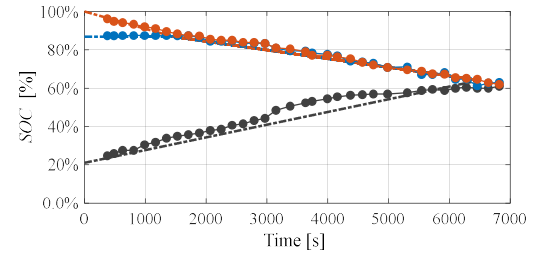


Fig. 16. The measured SOC profile during balancing operation of three cells. The balancing time follows (8)-(13) depending on each condition (the balancing current, load current, and initial SOC). Initial SOC distribution of (a) $SOC_3 > SOC_2 > SOC_1$, (b) $SOC_3 > SOC_1 > SOC_2$, and (c) $SOC_1 > SOC_2 > SOC_3$.

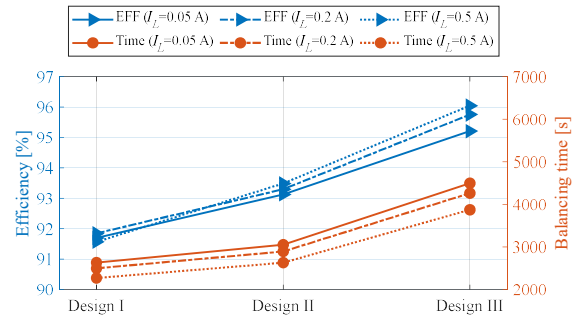


Fig. 17. The measured efficiency and balancing time versus various load currents, $I_L = 0.05$ A, 0.2 A, and 0.5 A. The initial distribution of SOC is as follows: $SOC_1 = 50\%$, $SOC_1 = 65\%$, and $SOC_1 = 100\%$. According to these experimental results, the selection of candidates based on the analysis is well validated.

process performed in the study is confirmed based on the quantitative differences in the experimental results. As discussed in [24], the burst mode operation has nearly constant efficiency over the entire range of the load current, and the burst mode for the proposed circuit is also validated by the experimental results.

TABLE II
COMPARISON WITH PREVIOUS WORKS

	[30]	[31]	[19]	[22]	THIS WORK
Balance topology	C2P (Time shared)	C2P, P2C	C2L	P2L or C2L	C2L2C
Converter type	Flyback with a single transformer	Flyback with dual transformer	Flyback with multiple transformers	Flyback with multi winding transformer	Single inductor type
Additional power conversion circuit	Required	Required	Not required	Not required	Not required
Overall size	Small	Large	Large	Medium	Small
Components	Transformer/inductor (for N cells)	1 T	2 T	N T	N -Multi winding T
	Switch / diode (for N cells)	$2N - 2$ (S) $2N - 1$ (D)	$4N + 2$ (S) 2 (D)	$2N$ (S)	$2N + 1$ (S) $4N$ (S)
Efficiency	88 % *	92 % *	93.4 %	85 %	96.05 %
Speed	Medium	Medium	Fast	Medium	Fast

In the case of the switch, a unidirectional switch is counted as one switch.

C2P – Cell-to-pack, P2C – Pack-to-cell, P2L – Pack-to-load, C2L – Cell-to-load, C2C – Cell-to-cell, and C2L2C – Cell-to-load-cell.

The efficiency with a star(*) refers to balancing operation only.

VII. DISCUSSION

The proposed topology is compared with previous studies of [19], [22], [30], and [31] in Table II, which shows the balancing topology, converter type, size, efficiency, and balancing speed. In [30], $2N - 2$ of switches and $2N - 1$ of diodes are used. Thus, the decrease in efficiency is severe due to large diode conduction loss. In [31], $4N + 2$ of switches and 2 of diodes are used with two transformers for step-up/down operation, which increase the size, area, and weight of the circuit. Additionally, the balancing speed is limited because of the pack type balancing method [2], [7]. In [19], a high degree of freedom in balancing operation with fast balancing speed is achieved by applying a bidirectional flyback converter on each cell stage. However, each stage needs a corresponding transformer, which dominates the size and weight of the system. Moreover, unlike other studies, each stage also requires output current sensing with A/D conversion for the current control method. This further increases the size and complexity of the controller. In [22], $2N + 1$ of switches with multi winding transformer are used for a medium size. The balancing speed is limited due to the P2L type balancing method, which relies on only discharging the load current at each battery cell.

Compared to other studies [19], [22], [30], and [31], this study provides many advantages. Due to the characteristics of the proposed topology, an additional power conversion circuit is not required. A single-inductor power converter topology with optimization based on in-depth analysis achieves a smaller size and higher efficiency. The increase in the number of switches due to bidirectional operation may increase the cost; however, the cost of the switch has been continuously decreasing recently. Consequently, the proposed circuit performs both battery balancing and power conversion (voltage regulation) with better performance in terms of efficiency and balancing characteristics compared to those in previous studies.

VIII. CONCLUSIONS

This paper presents a high-efficiency balancing circuit with voltage regulation for low-power mobile applications, which is advantageous for integration into battery packs. To optimize the proposed topology, the principle of operation, balancing characteristics, and loss models were derived. The trade-off relationship between balancing performance and efficiency was

analyzed and discussed. Then, three optimal design candidates were selected through the Pareto Front. This study focused on the quantitative analysis of C2L2C operation with a burst mode, which is distinguished from previous studies. To verify the analysis performed in this work, a GaN-based prototype board was implemented. The results show that the proposed circuit operates with high efficiency up to 96.05 % and a high-speed balancing operation of 2,270 seconds (37.8 minutes) for three series-connected battery cells (2,000 mAh each cell). The outstanding performance and high functionality can minimize system redundancy and achieve a high power density for battery-powered low-power mobile devices.

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